

HTML and CSS Basics for Web Scraping

Practical Session

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What is pickle in Python?

- **Pickle Module:** A Python module used to serialize and deserialize Python objects.
- **Serialization:** Converting a Python object into a byte stream to save to a file or transfer over a network.
- **Deserialization:** Converting the byte stream back into a Python object.
- **Use Case:** Save program state, machine learning models, or complex data structures.
- **File Type:** Typically stored with a `.pkl` or `.pickle` extension.

Importing Pickle Module:

- `import pickle`

Example: Saving Data with pickle

```
[15] import pickle
      # Filepath to save the pickle file
      pickle_file = "panchyat_episodes.pkl"

      # Save the `episodes` list to a pickle file
      with open(pickle_file, "wb") as file:
          pickle.dump(episodes, file)

      pickle_file
```

➞ 'panchyat_episodes.pkl'

Explanation:

- episodes: Python object to be saved.
- open(..., 'wb'): Opens the file in binary write mode.
- pickle.dump(): Serializes and writes the object to the file.
- The file episodes.pkl can later be loaded using pickle.load().

Assignment 1: Web Scraping

Objective: Scrape and analyze episode data from Wikipedia.

Task Steps:

- 1 Scrape episode data from:
`https://en.wikipedia.org/wiki/List\_of\_Panchayat\_episodes`
- 2 Use `requests` and `BeautifulSoup` to extract episode text.
- 3 Dump the scraped data into a file.
- 4 Reload the data from the saved file.
- 5 Perform basic text preprocessing, e.g. Stop word filtering.
- 6 Count word frequencies and display the top 10 most frequent words.
- 7 Generate a **word cloud** from the cleaned text.

Sample Solution: `hw1_scrape_panchyat_wikipedia.ipynb`

Web Scrapping

The screenshot shows a web browser window with the URL `en.wikipedia.org/wiki/List_of_Panchayat_episodes`. The page displays a table titled "List of Panchayat episodes" with columns: No., Title, Directed by, Written by, and Original release date. The table contains four rows of data. The browser's developer tools are open, showing the HTML structure of the table and the CSS styles applied to it.

No.	Title	Directed by	Written by	Original release date
1	"Gram Panchayat Phulera"	Deepak Kumar Mishra	Chandan Kumar	3 April 2020
2	"Bhootha Ped"	Deepak Kumar Mishra	Chandan Kumar	3 April 2020
3	"Chakke Wali Kursi"	Deepak Kumar Mishra	Chandan Kumar	3 April 2020
4	"Hamara Neta Kaisa Ho?"	Deepak Kumar Mishra	Chandan Kumar	3 April 2020

The developer tools show the HTML structure of the table, including the `<table>` element and its `<tbody>` and `<tr>` elements. The CSS styles are also visible, showing the table's dimensions and font settings.

Web Scraping with BeautifulSoup

- **Goal:** Extract episode details from a Wikipedia page.
- **Tools Used:**
 - `requests` — To fetch web content.
 - `BeautifulSoup` — To parse HTML and extract data.
- **Target:** HTML tables with class `wikiepisodetable` (refer screenshot in the previous slide).
- Data is converted into a list of dictionaries (key-value format).
- Implemented in function `save_episode_data()`

Overview of the Function `save_episode_data`

`save_episode_data()`

- Takes parsed HTML (`html_soup`) as input.
- Finds all tables that contain episode information.
- Extracts header (column titles) and data (episode info).
- Stores each episode as a dictionary.
- Returns a list of all episodes.

Sample Output from the Function

Each episode is saved like:

```
loaded_episodes[:2]
```

```
[{'No.': '1',  
  'Title': '"Gram Panchayat Phulera"',  
  'Directed by': 'Deepak Kumar Mishra',  
  'Written by': 'Chandan Kumar',  
  'Original release date': '3\xa0April\xa02020\xa0(2020-04-03)'},  
 {'No.': '\n\nAbhishek Tripathi is a fresh engineering graduate w  
shot as there is no other option left after graduating. He reache  
office assistant. When Abhishek reaches Phulera, the doors of the  
situation inside the office room turns him down. Abhishek decides  
Admission Test) and taking admission in one of the IIMs.'}]
```

Why useful?

- Saved as CSV (or try JSON also!!)
- Easy to search, analyze, or visualize episode data

Frequent Terms

```
print ("Most frequent 5 words")
top_5_words = get_top_n_words(cleaned_text, n=5)
print(top_5_words)

print ("Most frequent 10 words")
top_10_words = get_top_n_words(cleaned_text, n=10)
print(top_10_words)
```

Most frequent 5 words

[('abhishek', 74), ('kumar', 52), ('pradhan', 50), ('village', 26), ('mishrachandan', 24)]

Most frequent 10 words

[('abhishek', 74), ('kumar', 52), ('pradhan', 50), ('village', 26), ('mishrachandan', 24), ('vikas', 23), (

Word cloud

```
print ("Wordclouds of Titles and Description")
print wordcloud(str_titles + ' ' + str_desc)
```

➡ Wordclouds of Titles and Description



Assignment 2: Web Scraping

Objective: Scrape and analyze episode data from Wikipedia.

Similar to Assignment 1 Task Steps:

- 1 Scrape episode data from:
`https://en.wikipedia.org/wiki/List_of_Mayday_episodes`
- 2 Use `requests` and `BeautifulSoup` to extract episode text.
- 3 Dump the scraped data into a file.
- 4 Reload the data from the saved file.
- 5 Perform basic text preprocessing, e.g. Stop word filtering.
- 6 Count word frequencies and display the top 10 most frequent words.
- 7 Generate a **word cloud** from the cleaned text.

Highlight the difference with Assignment1!

Assignment1 Sample Solution:

`hw1_scrape_panchyat_wikipedia.ipynb`

Thank You!